SDSC 6008 Final Project Proposal

Rainwater Harvesting System Design in Hong Kong

Track B Computational Modelling | Instructor Dr. XZ

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1 Research Question and Hypotheses

We will study how roof area, tank capacity and first-flush diversion depth affect the share of non-potable demand met by rainwater in Hong Kong. Using observed HKO rainfall [1], we will simulate storage for one building and compare the three factors and their interactions. The model will show the benefit of extra storage and the water lost through diversion under our demand assumptions. It cannot determine a safe diversion depth from water volume alone.

H1: We expect larger tanks to meet more demand, with smaller gains as capacity increases. We do not assume that tank capacity has the largest effect. H2: More first-flush diversion should reduce or leave unchanged the water available for use. H3: We expect a positive interaction between roof area and capacity: a larger roof should be more useful when the tank can store the extra water instead of overflowing.

2 Factors Levels and Response Variable

Factor and definition	Low	Centre	High
A Roof area feeding one tank (m^2)	300	500	700
B Usable storage capacity (m^3)	5	12.5	20
C First-flush depth per event (mm)	0	0.75	1.5

The levels describe hypothetical systems, not measured buildings or recommended installation sizes. We chose 300-700 m^2 to compare smaller and larger collection areas while keeping demand fixed. At the assumed mean demand of 1.2 m^3/day, the 5-20 m^3 tanks hold about 4-17 days of demand. The 0-1.5 mm range tests how much supply is lost through first-flush diversion; 0 mm is a water-yield benchmark only. The pilot will check whether these ranges produce distinguishable responses. WSD includes first-flush removal in rainwater treatment [2], but these levels are not local standards and the model will not assess water quality or compliance.

Our main response is annual coverage = 100 × rainwater supplied / non-potable demand (%), using the annual totals of daily volumes. We will also record annual backup water and overflow (m^3), and the number of days needing backup. Since the original supply source is unspecified, coverage does not necessarily measure potable-water savings.

We define a rainfall event as consecutive days with at least 0.1 mm of rain, following at least one dry day. Each event starts with a diversion allowance of C mm. On each wet day, we divert the smaller of the day's rain depth and the remaining allowance, then carry any unused allowance forward. A dry day resets the event. This rule lets us use daily data, although it cannot identify actual storm boundaries.

After diversion, inflow = 0.8 × A × retained rainfall / 1,000 m^3, where 0.8 allows for assumed additional collection losses. The model supplies demand from the previous day's storage before adding today's inflow. Water above capacity B overflows. Backup water covers any shortfall without refilling the tank. Each run starts with an empty tank in the preceding calendar year for warm-up; we calculate responses only for the target year.

3 Factorial Design Blocking and Response Surface

We will use 2021, 2022 and 2023 as blocks, preserving the same HKO station's complete daily records. Annual rainfall totals and dry-spell lengths can affect coverage independently of the engineering settings, so year is a nuisance blocking variable. Each year includes every setting; no treatment effect is confounded with block. Settings 1-8 form the full 2^3 factorial. Six face-centred axial settings and one centre complete a central composite design [3]. Table entries are independent run counts, not execution order.

Setting	A m^2	B m^3	C mm	2021	2022	2023
1	300	5	0	10	10	10
2	300	5	1.5	10	10	10
3	300	20	0	10	10	10
4	300	20	1.5	10	10	10
5	700	5	0	10	10	10
6	700	5	1.5	10	10	10
7	700	20	0	10	10	10
8	700	20	1.5	10	10	10
9	300	12.5	0.75	10	10	10
10	700	12.5	0.75	10	10	10
11	500	5	0.75	10	10	10
12	500	20	0.75	10	10	10
13	500	12.5	0	10	10	10
14	500	12.5	1.5	10	10	10
15	500	12.5	0.75	10	10	10

The design has 15 settings $\times$ 3 years $\times$ 10 repeats = 450 main runs. In R, we will estimate factorial main effects and interactions, then fit a quadratic response surface with linear, pairwise-interaction and squared terms plus year effects. We will check residuals, lack of fit and whether effects vary by year. New demand draws at intermediate settings will test the fitted surface; 2024 rainfall provides a further weather check. If the surface fits poorly, we will report direct simulation results.

4 Randomization and Independent Replication

We will randomize the 150 runs within each year and save the scheduling seed. Separate recorded seeds generate independent daily demand draws, $D = 1.2 \times U$ m^3, $U \sim \text{Uniform}(0.8, 1.2)$, including warm-up. The mean and variation are assumptions. Ten annual repeats per setting and year provide an initial estimate of demand variability at a manageable computational cost, conditional on that year's rainfall. Days and identical reruns are not independent repeats. Before fixing the design, we will review pilot standard errors and increase the repeat count equally across cells if needed.

5 Feasibility Timeline and Planning

We will save a copy of the HKO CSV and use base R, without a physical experiment. Checks will cover dates, quality flags, water balance and storage limits; missing rainfall will not be set to zero. Sensitivity checks will vary collection losses, demand level and variation, event separation to two dry days, and daily timing. If code fails, we will fix it, rerun the affected seeds and record the changes. We will not cut repeats just to save time.

Week 7 covers the proposal and data preparation; Week 8 covers the pilot and final design. We will run the experiments and factorial analysis in Weeks 9-10, then fit and validate the response surface in Week 11. For Week 12, we will prepare the report, R appendix, joint five-minute video and individual reflections. We will keep the inputs, seeds, outputs and annotated screen recordings, and record each member's contribution.

References

[1] HKO. Daily Total Rainfall at the Hong Kong Observatory. Official CSV.

[2] WSD (2024). Technical Specifications on Grey Water Reuse and Rainwater Harvesting, 2nd ed., Sections 2.2-2.5.

[3] NIST/SEMATECH. e-Handbook of Statistical Methods, Section 5.3.3.6.1, Central Composite Designs.